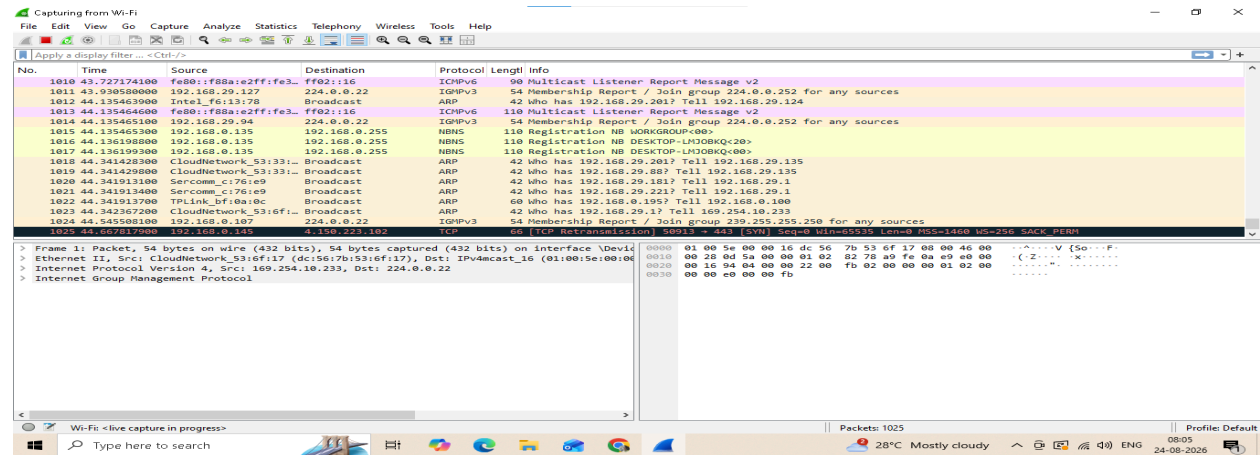


Practical 7: Web Security with SSL/TLS

AIM: Configure and implement secure web communication using SSL/TLS protocols, including certificate management and secure session establishment.

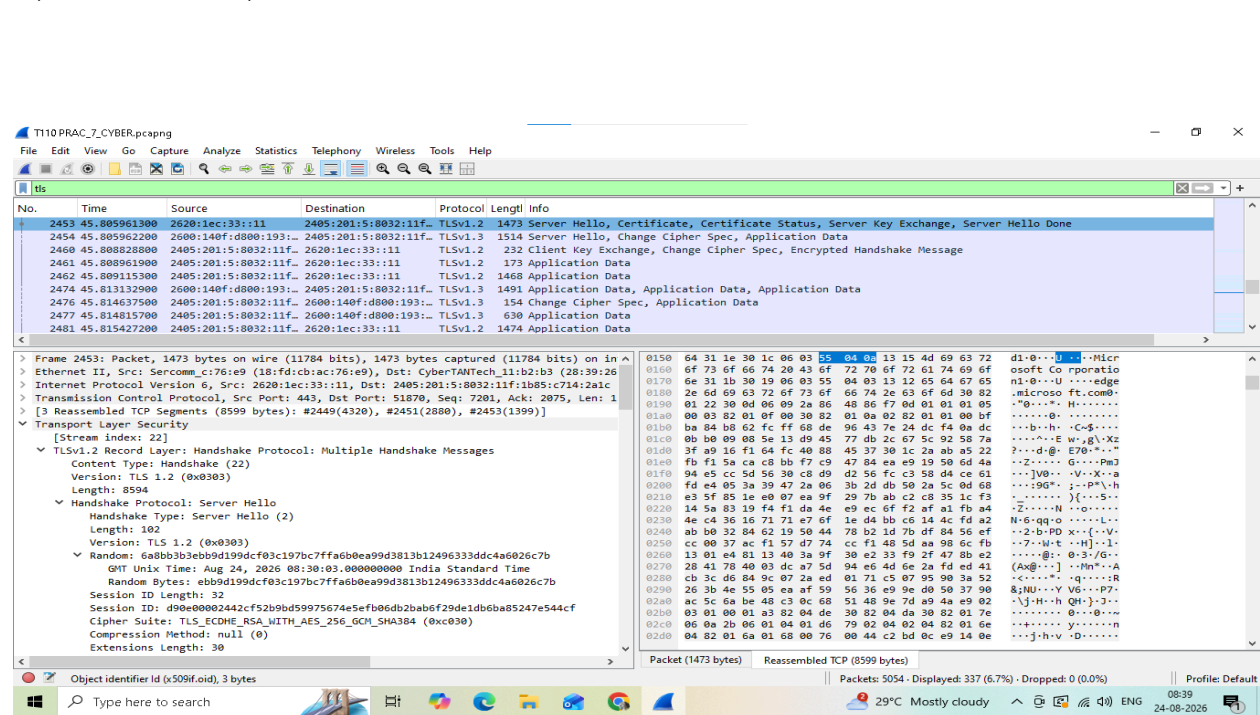
Step 1: Wireshark Open Apna active network interface select

Wi-Fi → **Wi-Fi**



Step 2: Browser mein HTTPS website open

Step 3: Wireshark mein TLS packets filter



Certificate

Practical 7: Web Security with SSL/TLS

The image shows a Wireshark capture of a TLS handshake. The packet list on the left shows the following packets:

- 2453 45.805961300 2620:1ec:33::11 2405:201:5:8032:11f... TLSv1.2 1473 Server Hello, Certificate, Certificate Status, Server Key Exchange, Server Hello Done
- 2454 45.805962200 2620:140f:d800:193... 2405:201:5:8032:11f... TLSv1.3 1514 Server Hello, Change Cipher Spec, Application Data
- 2460 45.808828800 2405:201:5:8032:11f... 2620:1ec:33::11 TLSv1.2 232 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
- 2461 45.808961900 2405:201:5:8032:11f... 2620:1ec:33::11 TLSv1.2 173 Application Data
- 2462 45.809115300 2405:201:5:8032:11f... 2620:1ec:33::11 TLSv1.2 1468 Application Data
- 2474 45.813132900 2620:140f:d800:193... 2405:201:5:8032:11f... TLSv1.3 1491 Application Data, Application Data, Application Data
- 2476 45.814637500 2405:201:5:8032:11f... 2620:140f:d800:193... TLSv1.3 154 Change Cipher Spec, Application Data
- 2477 45.814815700 2405:201:5:8032:11f... 2620:140f:d800:193... TLSv1.3 630 Application Data
- 2481 45.815427200 2405:201:5:8032:11f... 2620:1ec:33::11 TLSv1.2 1474 Application Data

The packet details pane shows the following information for the selected packet (1473):

- Object Id: 2.5.4.10 (id-at-organizationName)
- DirectoryString: printableString (1)
- printableString: Microsoft Corporation
- RDNSSequence item: 1 item (id-at-commonName-edge.microsoft.com)
- RelativeDistinguishedName item (id-at-commonName-edge.microsoft.com)
- Object Id: 2.5.4.3 (id-at-commonName)
- DirectoryString: printableString (1)
- printableString: edge.microsoft.com
- subjectPublicKeyInfo
- algorithm (rsaEncryption)
- Algorithm Id: 1.2.840.113549.1.1.1 (rsaEncryption)
- Padding: 0
- subjectPublicKey [...]: 3082010a0282010100bfb48b62fcff68de96437e24dcf40
- RSA Public Key
- modulus: 00b0fb48b62fcff68de96437e24dcf40adcb0b009085e13d94577c
- publicExponent: 65537
- extensions: 12 items
- Extension (SignedCertificateTimestampList)
- Extension (id-ms-application-certificate-policies)
- Extension (id-ms-certificate-template)
- Extension (id-pe-authorityInfoAccess)
- Extension (id-ce-subjectKeyIdentifier)
- Extension (id-ce-keyUsage)

The packet bytes pane shows the raw data of the packet, including the handshake message and the encrypted handshake message.

The image shows a Wireshark capture of a TLS handshake. The packet list on the left shows the following packets:

- 2453 45.805961300 2620:1ec:33::11 2405:201:5:8032:11f... TLSv1.2 1473 Server Hello, Certificate, Certificate Status, Server Key Exchange, Server Hello Done
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- 2481 45.815427200 2405:201:5:8032:11f... 2620:1ec:33::11 TLSv1.2 1474 Application Data

The packet details pane shows the following information for the selected packet (1473):

- Certificate Length: 1980
- Certificate [...]: 308207b308205a0e0030201020213330000000e42938f25dc51a99b000000e
- signedCertificate
- algorithmIdentifier (sha384withRSAEncryption)
- padding: 0
- encrypted [...]: 8186235256b3dd33ba832a5cee7dc7ed5e508a6852f2754513763764d746e
- Certificate Length: 1421
- to Certificate [...]: 3082050930820471a00302010202100b0c6b2c466917b04773c6474df0c83
- Handshake Protocol: Certificate Status (22)
- Handshake Type: Certificate Status (22)
- Certificate Status Type: OCSP (1)
- OCSP Response Length: 2341
- OCSP Response
- responseStatus: successful (0)
- responseBytes
- ResponseType Item: 1.3.6.1.5.5.7.48.1.1 (id-pkix-ocsp-basic)
- BasicOCSPResponse
- responderID: byKey (2)
- byKey: b1ccc645214537fd8175643c328e1abb30d9552d
- producedAt: Aug 21, 2026 07:06:22.000000000 India Standard Time
- responses: 1 item

The packet bytes pane shows the raw data of the packet, including the handshake message and the encrypted handshake message.